Velocity and position graphs

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect a position-time graph to velocity using slope.

SUCCESS CRITERIA - I can...

1. Read positions and elapsed times from a graph.
2. Find velocity from position change divided by time.
3. Explain the difference between speed and velocity.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “velocity” and “speed” to explain your answer. Sentence starters are optional.

velocity: rate of position change, with direction

speed: how fast something moves, without direction

slope: vertical change divided by horizontal change on a graph

01 Warm-up: bring back an earlier idea

Start at **3 m**, move to **10 m**, then finish at **5 m**.
1. **Draw the route**.
2. **Find distance**, displacement with direction and final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

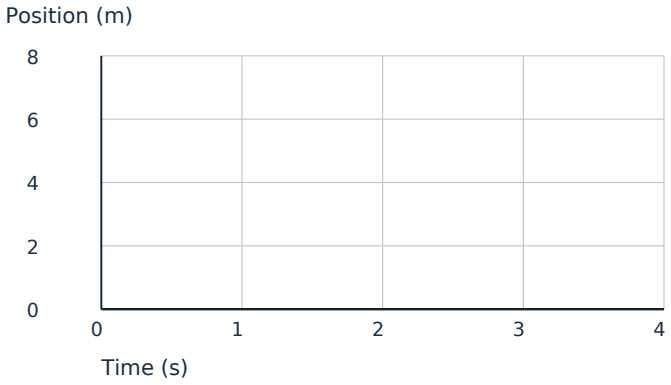
02 Worked example: explain the prediction

Reference: $v_{avg} = \Delta x/\Delta t$; average speed = distance/Δt.
Δ means change; Δt is elapsed time. Position-time slope gives velocity.

Worked reasoning: A cart moves from **1 m** to **7 m** in **3 s** at steady velocity. The position-time slope is (**7**-**1**)/**3**=**2 m/s**.

Your turn: 1. Use the blank axes to plot the cart's two measured positions and connect them.
2. **Draw a slope triangle**.
3. **Label its rise** in meters and its run in seconds.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

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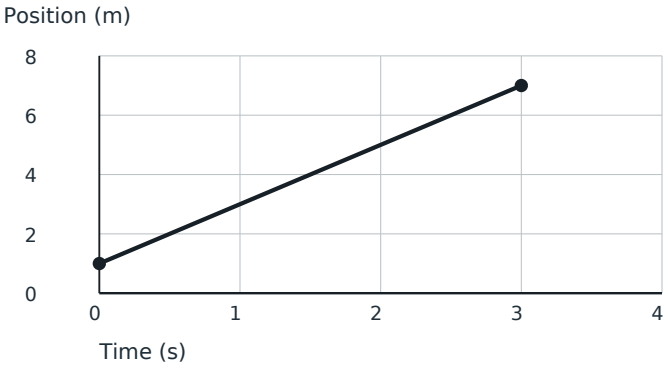
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read the motion picture

A motion sensor tracks a cart on a straight test track.
1. **Read its position** at **0 s** and **3 s**.
2. **Which direction** is it moving?
3. Use the changing position to **explain your answer**.

Sentence starter: I think … . I can point to …, which shows …, so … .



Marzano check - circle: ① ② ③

04 Find the rate

Use the graph in **3**. Position change = ____ - ____ = ____ m. Time = ____ s. Velocity = ____ m/s to the ____.

Sentence starter: I found … [unit]. This value means … in this situation.

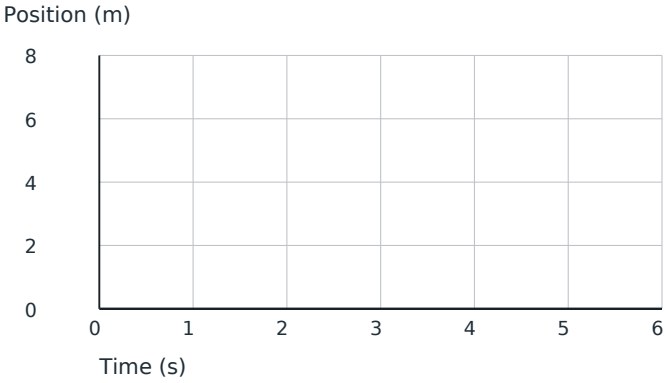
Marzano check - circle: ① ② ③

05 A pause

During another trial, the cart's position stays at **5 m** from **4 s** to **6 s**.
1. Sketch that interval on the axes.
2. Is the cart moving?
3. **Explain what stays unchanged**.

Support: The cart is ____ because its position ____.

Sentence starter: I think … . I can point to …, which shows …, so … .



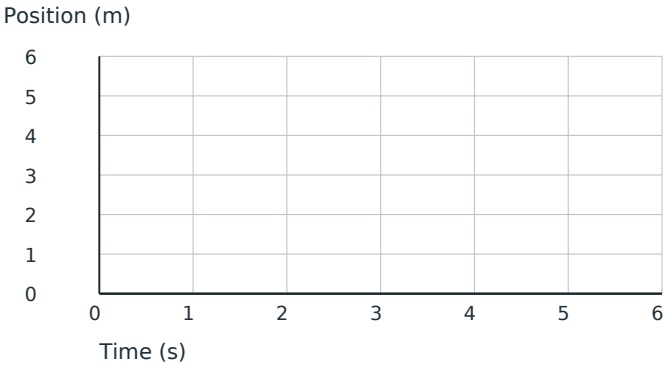
Marzano check - circle: ① ② ③

06 Draw a walk with a pause

A student walks along a marked hallway. Plot (time, position): (**0 s**, **1 m**), (**2 s**, **5 m**), (**4 s**, **5 m**), (**6 s**, **1 m**). Connect neighboring points with straight lines. **Label each interval**: moving right, stopped, or moving left.

Support: Data: t = **0**, **2**, **4**, **6 s**; x = **1**, **5**, **5**, **1 m**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Distance or displacement?

For the walk in **6**, find distance traveled and displacement. Why is average velocity zero even though the person walked?

Sentence starter: I think … . I can point to …, which shows …, so … .

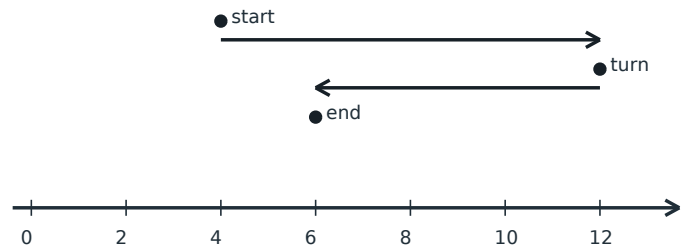
Marzano check - circle: 1 2 3

08 Review: connect an earlier idea

Start at **4 m**, move to **12 m**, then finish at **6 m**.
1. **Draw the route**.
2. **Find distance**, displacement with direction and final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found … [unit]. This value means … in this situation.



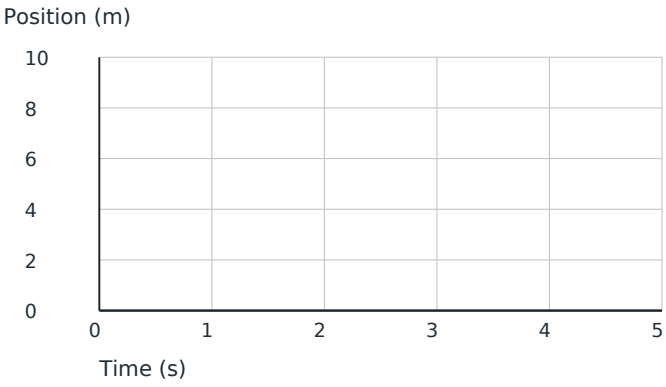
Position (m); right is positive

Marzano check - circle: 1 2 3

09 Exit check

A remote-control car moves steadily from the **9 m** mark to the **3 m** mark in **5 s**. **Find its velocity**, **including direction**, and **draw its position-time graph**.

Sentence starter: I found … [unit]. This value means … in this situation.

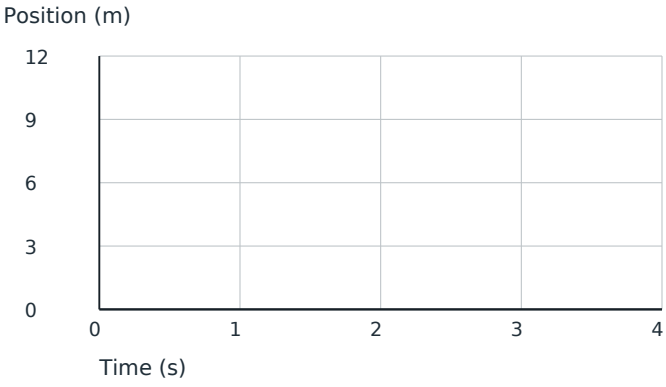


Marzano check - circle: 1 2 3

10 STRETCH: test the prediction

1. **Draw two straight position-time graphs** with different starting positions but the same velocity of **2 m/s**.
2. **Explain the feature they share**.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3